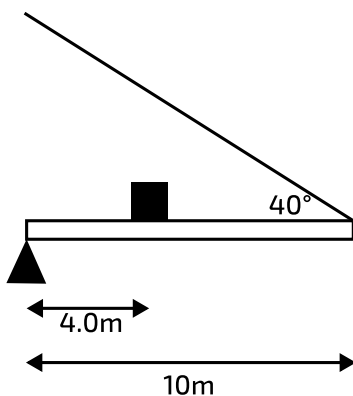


## B5 Moments

Strength of Earth's gravity at ground level =  $9.81 \text{ N kg}^{-1}$ . 1 tonne = 1000 kg.

As throughout this book, numeric answers should contain units. Where forces are asked for, ensure that the direction is in the answer (e.g. up/down). Assume that the mass is evenly distributed in the rulers, poles, planks, bridge spans mentioned in the questions.

- B5.1 A metre rule is pivoted about the 50 cm mark (which is the position of its centre of mass). Find the direction and magnitude of force  $F$ , acting at the 40 cm mark, needed to balance the rule if there is a downward 10 N force acting at the 90 cm mark.
- B5.2 A metre stick has its centre of mass at the 50 cm mark. A 150 g mass is stuck to the 5 cm mark with massless glue. The ruler balances when supported below the 29 cm mark. What is the mass of the ruler?
- B5.3 As shown in the diagram below, an 8.0 tonne steel block has been left on a bridge, 4.0 m from the left hand edge. The bridge beam has a mass of 16 tonnes. It is supported from below at the left edge, and the right edge is supported by a steel cable, which makes an angle of  $40^\circ$  to the horizontal. Calculate the tension in the cable. You may neglect the mass of the cable.



## CHAPTER B. MECHANICS

### Answers

**B5.1**  $F \times 10 \text{ cm} = 10 \text{ N} \times 40 \text{ cm}$   
 $F = 40 \text{ N}$

**B5.2**  $150 \text{ g} \times 24 \text{ cm} = m \times 21 \text{ cm}$   
 $m = 171 \text{ g}$

**B5.3**  $8000 \text{ g} \times 4 + 16000 \text{ g} \times 5 = 10 \times T \sin 40^\circ$   
 $T = 171 \text{ kN}$